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Average Numbers of Scintillation Photons and Electrons Produced by an Alpha Particle in High-Density Xenon Gas

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The average numbers of scintillation photons and electrons produced by an α -particle with 5.49 MeV were determined in high-density xenon gas in the density range of 0.03–0.12 g/cm³. No significant density dependence was observed for the number of excited atoms or liberated electrons. The average energies expended per excited atom (W_{ex} -value) and liberated electron (W -value) were found to be 34.1 ± 2.4 and 20.9 ± 0.4 eV, respectively. It was also found that some liberated electrons escape from recombination at zero electric field.

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1. Introduction

Xenon is suitable as a γ -ray detection medium because of its large atomic number, high ionization yield, and high resistance to radiation damage. Xenon is also excellent as a detection medium because both signals of liberated electrons and scintillations are simultaneously available. The charge collection triggered by scintillation photons allows us to construct a three-dimensional time projection chamber. The time projection chamber using xenon gas (Xe-TPC) is expected to be a sophisticated γ -ray camera with a high angular resolution in the MeV energy range.^{1–3} Its disadvantage is its low detection efficiency owing to its low density under normal condition. A Xe-TPC operated at a high density is one of approaches to overcoming this disadvantage.

There are two components in the scintillation of gaseous xenon, i.e., excitation luminescence and recombination luminescence, which are described in §2. Under a complete collection of liberated electrons, the scintillation process of recombination luminescence is prevented. Thus, a Xe-TPC obtains the timing of γ -ray incidence using scintillation photons originating from excitation luminescence, whose number is normally equated with the number of excited atoms directly produced by a charged particle.

In order to design a Xe-TPC, it is necessary to understand how many excited atoms and electrons are produced by a charged particle in xenon gas above 0.06 g/cm³. However, there is no report concerning the average number of excited atoms produced by a charged particle in pure xenon gas above 0.06 g/cm³. Furthermore, there is no reliable average energy expended per excited atom in gaseous xenon even at 0.005 g/cm³ (1 atm), owing to the inconsistent values reported in previous works cited below.

The absolute scintillation yields produced by α -particles were measured by Miyajima *et al.*⁴ and Saito *et al.*⁵ These absolute scintillation yields include both excitation and recombination luminescences. As a result, the numbers of excited atoms have been estimated from the total scintillation yields by measuring the ratio of the scintillation yield originating from the excitation luminescence to total scintillation yield.^{6,7} Saito *et al.*⁷ reported that the average energy expended per excited atom at 0.005 g/cm³ is estimated from the average energy expended per scintillation photon obtained by Miyajima *et al.*⁴ to be 59.4 ± 2.4 eV. In contrast, Saito *et al.*⁶ reported that the average number of

excited atoms in the density range of 0.005–0.056 g/cm³ (1–10 atm) is estimated from their own results⁵ to be 1.59×10^5 for 5.49 MeV α -particles, which corresponds to 34.5 eV in the average energy expended per excited atom. The average energy expended per excited atom obtained by Saito *et al.*⁷ is 1.7 times higher than that obtained by Saito *et al.*⁶

The absolute scintillation yield produced by 5.9 keV X-rays has been recently reported by Carmo *et al.*⁸ They reported that the average energy expended per excited atom is estimated to be 111 ± 16 eV from the ratio of the primary scintillation yield to the secondary scintillation yield in xenon gas at 0.006 g/cm³.⁸ Besides, the average energy expended per excited atom for 59.6 keV γ -rays in Xe/He (90/10) at 15 atm was estimated by Parsons *et al.*⁹ to be 76 ± 12 eV. The result was equated with that in pure xenon gas.⁹ We expect that the average energies expended per excited atom in gaseous xenon for X-rays, γ -rays or electrons are almost equal to that for α -particles. However, their results^{8,9} are inconsistent with those for α -particles.^{6,7}

The inconsistent reports^{6–9} on absolute scintillation yield in xenon gas and its lack in a high-density region encouraged us to carry out this work. α -Particles in the MeV region produce many scintillation photons in the vicinity of the α -ray source in xenon gas because of the short range. As a result, the α -ray source becomes a highly luminous, fixed, and nearly point light source. Taking account of its advantage, the absolute scintillation yields produced by 5.49 MeV α -particles were measured in pure xenon gas in the density range of 0.03–0.12 g/cm³. In order to improve the reliability of absolute measurement, the absolute ionization yields were simultaneously measured.

In this study, the average energies expended per excited atom and liberated electron in high-density xenon gas up to 0.12 g/cm³ were determined and discussed. The density dependence of total scintillation yield including both excitation and recombination luminescences was also discussed.

2. Excitation and Recombination Luminescence

According to Platzman,¹⁰ the energy balance for the kinetic energy E_c of a charged particle absorbed in a gas is given by

$$E_c = N_i E_i + N_{\text{ex}} E_{\text{ex}} + N_i \epsilon_{\text{h}}, \quad (1)$$

where N_i denotes the average number of electrons produced by a charged particle, E_i the average ionization potential of

the gas to form a liberated electron, N_{ex} the average number of excited atoms produced by a charged particle, E_{ex} the average excitation potential of the gas to form an excited atom, and ϵ_h the average kinetic energy of sub-excitation electrons. W -value is the average energy expended per liberated electron and defined as

$$W = \frac{E_c}{N_i}. \quad (2)$$

Analogously, W_{ex} -value is defined as

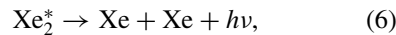
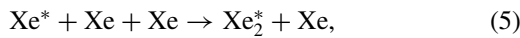
$$W_{\text{ex}} = \frac{E_c}{N_{\text{ex}}}, \quad (3)$$

and denotes the average energy expended per excited atom.

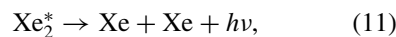
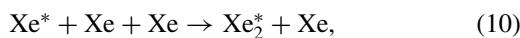
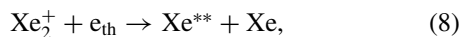
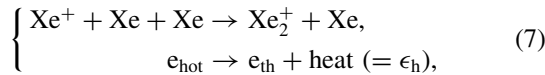
W -value can be directly obtained by measuring the number of electrons liberated in an ionization chamber. In contrast, we have to indirectly estimate W_{ex} -value from scintillation yields caused by excited atoms. Scintillation yield has been characterized often by W_s -value defined as the average energy expended per scintillation photon. W_s -value is given by

$$W_s = \frac{E_c}{N_s}, \quad (4)$$

where N_s is the average number of scintillation photons produced by a charged particle. W_s -value is not same as W_{ex} -value because there are two origins of scintillation in xenon gas. Firstly, an excited atom emits a scintillation photon above 1 atm as follows:¹¹⁾



where Xe is atomic xenon, Xe^* is an excited xenon atom, Xe_2^* is an excited xenon molecule, and $h\nu$ is a scintillation photon. Secondly, the scintillation photon is also emitted as follows:¹¹⁾



where Xe^+ denotes an atomic xenon ion, Xe_2^+ a molecular xenon ion, Xe^{**} a highly excited xenon atom, e_{hot} a sub-excitation electron, and e_{th} a thermalized electron. The scintillations originating from the former process is called excitation luminescence. Those originating from the latter process is called recombination luminescence. Both of the scintillations due to excitation and recombination luminescences have a wavelength of 175 nm and a full width at the half maximum (FWHM) of about 15 nm.⁵⁾

One of the differences between excitation and recombination luminescences is in the electric field dependence. In process (8), $\text{Xe}_2^+ - e_{\text{th}}$ pairs are coupled by coulomb force depending on the initial separation between Xe_2^+ and e_{th} . By applying a high electric field that overcomes the coulomb force, recombination is prevented. Since one excited atom emits one scintillation photon,⁶⁾ this means that N_s at a sufficiently high electric field is equal to N_{ex} .

Another difference is in the time evolution. The time evolution of excitation luminescence is mainly determined by the spontaneous decay of Xe_2^* , whose lifetimes are 4.6 ns in a singlet and 99 ns in a triplet,¹²⁾ shown in process (6). In contrast, the time evolution of recombination luminescence is strongly influenced by the reaction rate of recombination shown in process (8).¹¹⁾ In our previous work,¹³⁾ we tried to collect scintillation photons originating from recombination luminescence using a main amplifier at a shaping time constant of 300 μs . However, a complete collection of recombination luminescence could not be achieved below 0.09 g/cm³. This means that the luminous duration is considerably long at a low density.

3. Experimental Methods

3.1 Apparatus and gas-filling system

The cross-sectional view of the apparatus is shown in Fig. 1 for the measurements of the average numbers of scintillation photons and liberated electrons. The chamber consisted of a cathode, a gridded anode, and a viewport. An α -ray source of ²⁴¹Am, electrochemically deposited in a circle 7 mm in diameter at the center of a stainless steel disk, was fixed on the cathode. A collimator 0.1 mm in thickness and 2.0 mm in diameter covers the α -ray source to reduce count rate. The cathode was connected to a well-stabilized high-voltage power supply through a noise filter. The power supply can provide an electric field between the cathode and the anode. The center part of the anode with a diameter of 24 mm was gridded in order to make it transparent for scintillation photons and to simultaneously collect liberated electrons. The grid wires were 0.1 mm in diameter and were arrayed at 1.0 mm intervals. Charge signals can be completely collected by the anode and grid wires. On the other hand, scintillation photons transmit through the gridded anode with a transparency of 90%. The viewport made of MgF₂ 10 mm in thickness was used to transmit 175 nm scintillation photons outside the chamber. A vacuum-ultraviolet sensitive photomultiplier tube (PMT; Hamamatsu Photonics R6836)

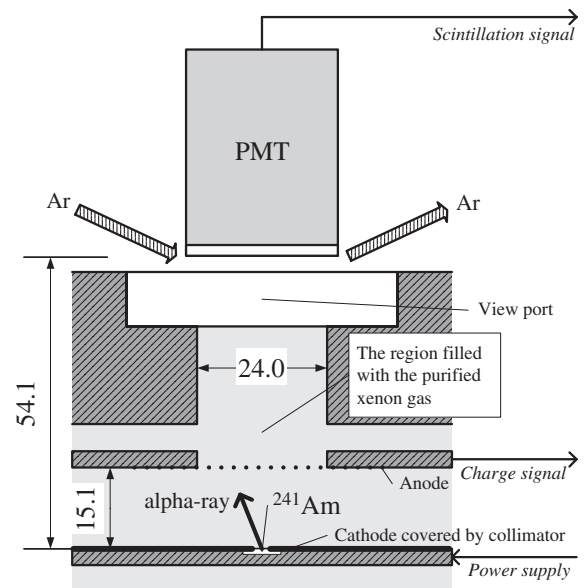


Fig. 1. Cross-sectional view of an experimental apparatus, with dimensions in mm.

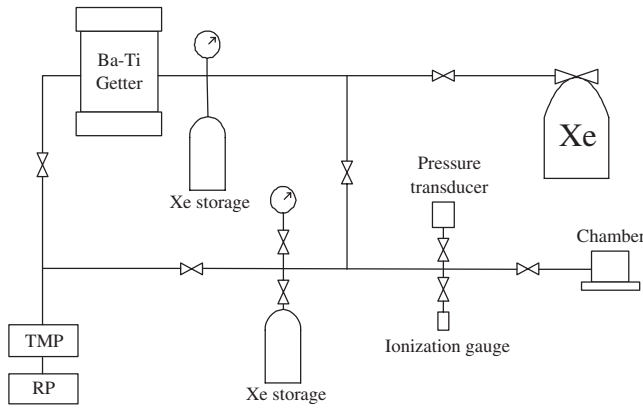


Fig. 2. Block diagram of purification and gas evacuation system. TMP and RP denote the turbo-molecular and rotary pumps, respectively.

was installed to observe scintillation photons through the viewport. The distance between the α -ray source and the window of PMT was 54.1 ± 0.2 mm. Ar gas was kept flowing through the gap between the viewport and the window of PMT to prevent light attenuation.

The chamber was connected to a vacuum and gas-filling system with a xenon gas purifier, as shown in Fig. 2. The gas-filling system and chamber was heated and evacuated to approximately 1.0×10^{-5} Pa. Xenon gas, whose initial purity was 99.999% or higher, was purified for 72 h with Ba-Ti getters¹⁴ heated at 600 °C. Then, the chamber was filled with purified xenon gas. The purity was checked by measuring both the electron lifetime and drift velocity of liberated electrons. The lifetime was sufficiently long for our measurement and longer than 1 ms at 0.117 g/cm³. The drift velocity at 0.063 g/cm³ was well consistent with that in pure xenon gas¹⁵ in a reduced electric field E/N ranging from 0.04 to 0.4 Td, where E is the electric field, N is the number density, and $1 \text{ Td} = 1 \times 10^{-17} \text{ V cm}^2$. Gas pressure was monitored by a pressure transducer (MKS 890B13P).

3.2 Signal processing for scintillation photons

The PMT was biased at 601 ± 1 V for more than 3 h before measurements for the stabilization of operation. The output pulses from the anode of PMT were sent to a charge sensitive preamplifier with a decay time constant of 4.4 ms, and then fed to the main amplifier (Clear Pulse CP4016) with a shaping time constant (STC) of 50 μ s, which achieves a complete collection of excitation luminescence. The pulse heights of the shaped signals were analyzed with a multi-channel analyzer (MCA; Clear Pulse CP1114A).

Figure 3 shows typical pulse height distributions of scintillation signals without and with an electric field. The peak region was fitted to a gaussian to determine average pulse height. Pulse height was converted to the number of electrons collected by the anode of PMT, N_{pmt} , using a precision pulse generator (Ortec 419). The pulse generator had been calibrated with an α -ray source composed of ^{237}Np , ^{241}Am , and ^{247}Cm using a windowless PIN photodiode with a thin dead layer,¹⁶ assuming that the ϵ -value of Si is 3.62 eV¹⁷ at room temperature. The number of scintillation photons, N_p , was determined using

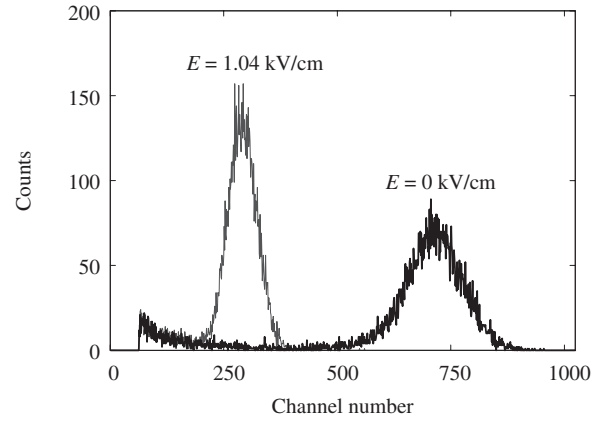


Fig. 3. Pulse height distributions of scintillation signals in xenon gas at 0.055 g/cm³ without (thick line) and with (fine line) an electric field of 1.04 kV/cm.

$$N_p = \frac{N_{\text{pmt}}}{Q_{\text{pmt}} G_{\text{pmt}} \Omega_{\text{pmt}}}, \quad (12)$$

where G_{pmt} denotes the gain of PMT at the operating voltage, Q_{pmt} the quantum efficiency of the photocathode of PMT for xenon scintillation photons, and Ω_{pmt} the collection efficiency of scintillation photons at the window of PMT within the collection time.

The spectral quantum efficiency of the photocathode, q_e , was measured as a function of wavelength λ by the manufacturer (Hamamatsu Photonics) for a normal incidence as shown in Fig. 4. Q_{pmt} is given by

$$Q_{\text{pmt}} = \frac{\int I(\lambda) q_e(\lambda) d\lambda}{\int I(\lambda) d\lambda}, \quad (13)$$

where $I(\lambda)$ is the relative intensity of scintillation photons at λ , i.e., spectral intensity. Q_{pmt} was estimated to be 9.6% from known $I(\lambda)$.⁵⁾ G_{pmt} was measured to be 8.31×10^4 by the manufacturer. The gain includes a collection efficiency of photoelectrons at the first dynode of PMT.¹⁸⁾

Ω_{pmt} corresponds to the average probability of one scintillation photon hitting the window of PMT. The probability is affected by the emission angle of α -particles. In order to determine Ω_{pmt} , Monte Carlo simulation for photon transport in the geometric framework of the chamber was performed. The following assumption and experimental results were applied in the estimation of Ω_{pmt} :

- (1) ^{241}Am is uniformly distributed on the source and isotropically emits α -particles of 5.49 MeV.
- (2) Scintillation photons are produced along the straight α -particle track.
- (3) The scintillation yield per unit length is proportional to the stopping power dE/dx of α -particle in the xenon gas. dE/dx was calculated using the SRIM code.¹⁹⁾
- (4) Scintillation photons are emitted isotropically.
- (5) The gridded anode and surrounding walls made of stainless steel absorb scintillation photons completely. According to Saito *et al.*,⁷⁾ the reflectance of stainless steel for xenon scintillation photons is negligible ($\leq 3\%$).

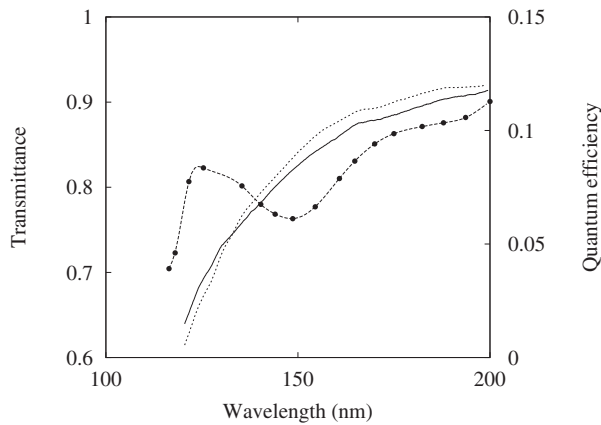


Fig. 4. Spectral quantum efficiency of the photocathode of PMT and spectral transmittance of the viewport as a function of wavelength. The broken line with measurement points (●) shows the quantum efficiencies. The solid line shows the transmittances. We also show the transmittances of viewport used in our previous work (dashed line).²²⁾

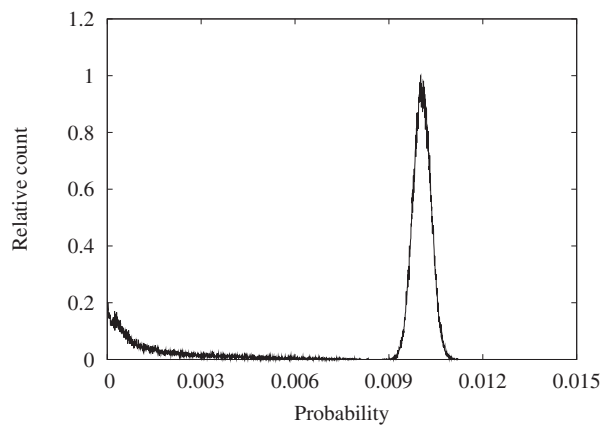


Fig. 5. Distribution of probability of one scintillation photon hitting the window of PMT calculated at 0.055 g/cm³.

- (6) When photons hit the viewport, they are in accordance with the law of Snell with a refraction index of 1.44 ± 0.01 .²⁰⁾
- (7) The spectral transmittance of the viewport, t_{win} , as a function of λ was measured by the manufacturer (Ohyo Koken Kogyo) for a normal incident angle, as shown in Fig. 4. The transmittance for xenon scintillation photons was estimated using eq. (13) to be 0.885, substituting $t_{\text{win}}(\lambda)$ for $q_e(\lambda)$.

We calculated the distribution of the probability of one scintillation photon hitting the window of PMT for isotropic α -ray emissions. The typical probability distribution calculated at 0.055 g/cm³ is shown in Fig. 5. The left-hand tail of the peak region in the spectrum, shown in Fig. 3, indicates events of an α -particle hitting the collimator. Ω_{pmt} was estimated from the average probability in the peak region.

3.3 Signal processing for collected charge due to ionization

Liberated electrons were collected by the gridded anode. The signals were sent to a charge sensitive preamplifier with

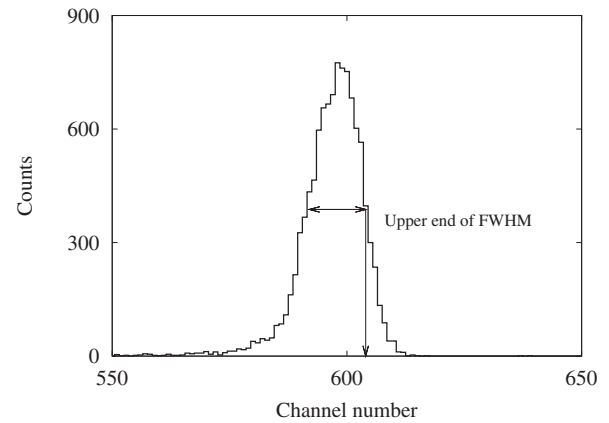


Fig. 6. Pulse height distribution of signals induced by liberated electrons in xenon gas at 0.055 g/cm³ and an electric field of 1.04 kV/cm. The channel at the upper end of FWHM is represented by the down-pointing arrow.

a decay time constant of 2.35 ms. Then, the output pulses were shaped by the main amplifier with a STC of 100 μ s, which achieves a complete collection of liberated electrons. The shaped signals were analyzed with the MCA. A typical pulse height distribution of charge signals is shown in Fig. 6. Since there is no Frisch grid in the chamber, the shape of the peak region is not Gaussian. As shown in Fig. 6, the channel at the upper end of FWHM was assumed to be proportional to average pulse height. Pulse height was converted to the number of electrons collected by the gridded anode, N_e , using the well calibrated pulse generator. When recombination is completely prevented, $N_e = N_i$.

3.4 Applied electric field

Experimentally, W_{ex} -value and W -value are respectively estimated from N_p and N_e at a sufficiently high electric field.

In our previous work,¹³⁾ the electric-field dependences of N_p and N_e have been studied relatively. Our previous results indicate that both N_p and N_e show fairly similar curves as a function of E/N without depending on density. It was also found that N_p and N_e are almost saturated at ~ 0.05 Td, and then, above 0.05 Td, N_p slightly decreases with E/N and N_e slightly increases with E/N . This is caused by the recombination unremoved by the electric field.

From a viewpoint of the complete removal of recombination, a higher electric field is better. However, there is an experimental limitation owing to discharge. We performed the absolute measurements of N_p and N_e at 0.4 Td.

The sufficiency of 0.4 Td should be evaluated. According to Jaffé,²¹⁾ $1/N_p$ and $1/N_e$ are proportional to $1/E$. We can extrapolate N_p and N_e at ∞ Td on the Jaffé model from the electric field dependences. The extrapolation values will become one of benchmarks for the sufficiency of E/N .

4. Results

4.1 Scintillation yield

We measured the number of scintillation photons, N_p , in xenon gas at various densities between 0.03 and 0.12 g/cm³ at 0.4 Td. The measurement of N_p was also performed at zero electric field in the density range of 0.03–0.12 g/cm³. The results are plotted as E_c/N_p in Figs. 7 and 8 with the

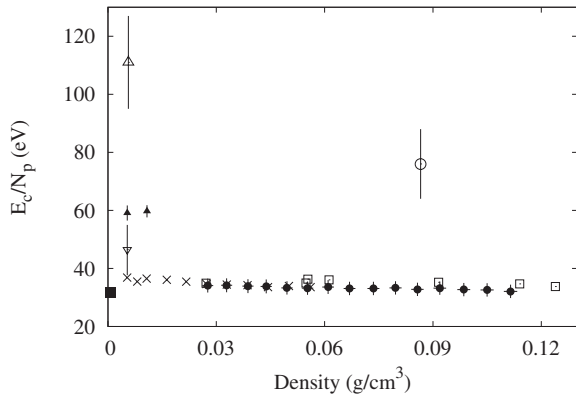


Fig. 7. Density dependence of scintillation yield in xenon gas at a high electric field. Filled circles (●) are the present data obtained at 0.4 Td (1 Td = 1×10^{-17} V cm²). Open squares (□) are our data at ∞ Td on the Jaffé model. Also shown are previous results by other groups: Parsons *et al.*⁹ (○), Saito *et al.*⁶ (×), Saito *et al.*⁷ (▲), Carmo *et al.*⁸ (△), and Date *et al.*²⁴ (■). The previous result of Miyajima *et al.*⁴ was reanalyzed and plotted (∇).

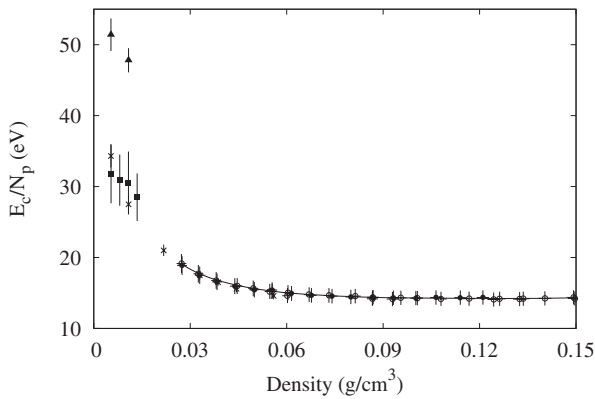


Fig. 8. Density dependence of scintillation yield in xenon gas at zero electric field. Filled circles (●) with a solid line are the present data. Open circles (○) are the previous data obtained using another viewport.²² Also shown are absolute measurements by other groups: Miyajima *et al.*^{4,7} (▲), Saito *et al.*⁵ (×), and Nemoto *et al.*²³ (■). Note that the scintillation yields of Nemoto *et al.*²³ were obtained assuming that $W_s = 17.9 \pm 1.5$ eV³¹ in liquid xenon.

results of previous studies.^{4-9,22-24} The pressures measured in the previous studies were converted to densities at 300 K.

The experimental errors concerning E_c/N_p are evaluated as follows. The error of the peak position in the spectrum is 0.1%. The error in the charge calibration is 0.7%. The error of G_{pmt} is estimated from the reproducibility to be 1.5%. The error of Ω_{pmt} is estimated from the comparison between the present data and the previous data,²² which was measured with an almost similar experimental apparatus except for the viewport, to be typically 2.1%. The error concerning $q_e(\lambda)$ of Q_{pmt} is 5.4%. The error concerning $I(\lambda)$ of Q_{pmt} is 3.6%, assuming that the error of peak of $I(\lambda)$ (175 nm) is ± 3 nm. The total error of N_p is 7.0%.

The error of Ω_{pmt} induced by the reflectance of stainless steel is neglected. This is expected to be less than 3%, assuming that its reflectance is less than 3%.

4.2 Ionization yield

We performed measurements for the number of liberated

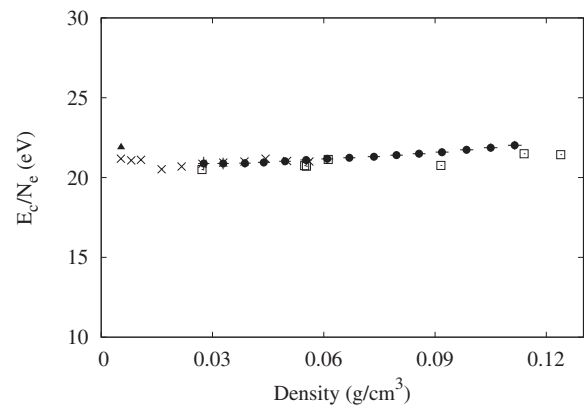


Fig. 9. Density dependence of ionization yield in xenon gas. Filled circles (●) are the present data obtained at 0.4 Td. Open squares (□) are our data at ∞ Td on the basis of the Jaffé model. Also shown are measurements by other groups: Jesse and Sadauskis²⁵ (▲) and Saito *et al.*⁶ (×).

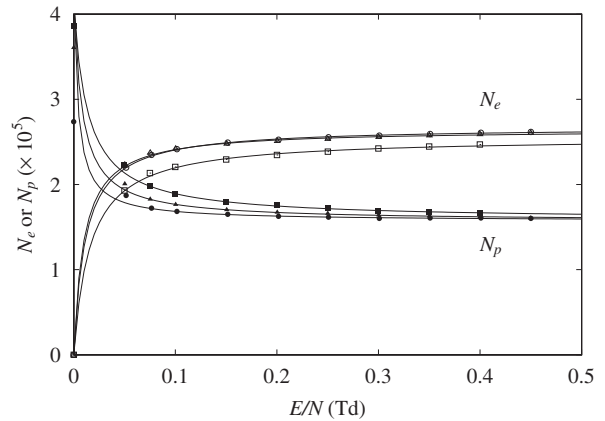


Fig. 10. Electric-field dependences of number of liberated electrons, N_e , and number of scintillation photons, N_p , in xenon gas at various densities. Filled points show N_p ; 0.027 (●), 0.055 (▲), and 0.114 g/cm³ (■). Open points show N_e ; 0.027 (○), 0.055 (△), and 0.114 g/cm³ (□). Solid lines represent the fitted curves on the basis of the Jaffé model.

electrons, N_e , at the same densities as that used in the measurement of N_p at 0.4 Td. The results are plotted as E_c/N_e in Fig. 9 with the previous results.^{6,25} The pressures measured in the previous studies were converted to densities at 300 K.

The experimental errors concerning E_c/N_e are evaluated as follows. The errors of average pulse height are 0.8–1.9%. The error in the charge calibration is 0.7%. The total error of N_e is 1.0–2.0%.

4.3 Electric-field dependence

We measured the electric-field dependences of N_p and N_e at densities of 0.027, 0.055, and 0.114 g/cm³. The results are plotted in Fig. 10 with the fitted curves of the Jaffé model. It is shown in Fig. 10 that the results are well fitted to the Jaffé model in a reduced electric field range of 0.1–0.45 Td. Based on the Jaffé model, we extrapolated E_c/N_e and E_c/N_p at ∞ Td using both present and previous results,¹³ which were calibrated with the present absolute results. The extrapolation values are plotted in Figs. 7 and 9.

5. Discussion

5.1 W_{ex} -value and W -value

At 0.4 Td, E_c/N_p demonstrates a slight decrease with increasing xenon density as shown in Fig. 7, whereas E_c/N_e shows a slight increase as shown in Fig. 9. These tendencies of E_c/N_p and E_c/N_e seem to be attributed to the incomplete suppression of recombination at a high density. As shown in Fig. 10, the changes in both N_p and N_e between 0.05 and 0.4 Td at 0.114 g/cm³ are larger than those at 0.027 g/cm³. This indicates an increase in the number of $\text{Xe}_2^+-e_{\text{th}}$ pairs strongly coupled by coulomb force in process (8) at 0.114 g/cm³. The increase in the number of these $\text{Xe}_2^+-e_{\text{th}}$ pairs can be explained in terms of the decrease in the initial separation between Xe_2^+ and e_{th} because the initial density of $\text{Xe}_2^+-e_{\text{th}}$ pairs increases with xenon density.²⁶⁾ The density dependences of E_c/N_p and E_c/N_e imply that the number of unremoved scintillation photons originating from the recombination luminescence at 0.4 Td increases with density owing to the increase in the number of strongly coupled $\text{Xe}_2^+-e_{\text{th}}$ pairs. Thus, W_{ex} -value and W -value are probably independent of xenon density in the investigated density range.

The extrapolation values at ∞ Td on the Jaffé model support this prediction. The results of E_c/N_p and E_c/N_e seem to be roughly independent of xenon density. The average E_c/N_p and average E_c/N_e at ∞ Td are 35.1 and 21.0 eV, respectively. At the lowest density of 0.028 g/cm³, the results of E_c/N_p and E_c/N_e at 0.4 Td are consistent with the above extrapolations.

The prediction is also supported by the electric-field dependence and density dependence of $E_c/(N_p + N_e)$ shown in Figs. 11 and 12, respectively. It is shown in Fig. 11 that there is no dependence of $E_c/(N_p + N_e)$ on E/N above 0.05 Td. This proves that one recombined $\text{Xe}_2^+-e_{\text{th}}$ pair emits one scintillation photon. Thus, $N_p + N_e$ is a quantity independent of the recombination luminescence not removed by an electric field. As shown in Fig. 12, there is no observable density dependence of $E_c/(N_p + N_e)$.

Two origins for the density dependences of N_{ex} and N_i have been pointed out. The first is the quenching of scintillation by collisions between excited states or excited dimers.^{27,28)} Two excited atoms or dimers convert to one electron through the quenching process. If the quenching process is dominant in scintillation processes, the density increase results in an increase in $E_c/(N_p + N_e)$ since the reaction rate of quenching increases with density. Our results of $E_c/(N_p + N_e)$ show that the scintillations are almost unquenched in the investigated density region. The second is the formation of a local conduction band structure in xenon gas, i.e., xenon clusters.^{29,30)} The formation results in a decrease in W -value. Although our results of E_c/N_e seem to be affected by the unremoved recombination, no downward trend of E_c/N_e is observed.

Taking the foregoing discussions into account, we determined W_{ex} -value and W -value from the measurement at the lowest density of 0.028 g/cm³ to be 34.1 ± 2.4 and 20.9 ± 0.4 eV, respectively. N_{ex}/N_i is calculated to be 0.61 ± 0.04 . We summarized the results in Table I with those of previous studies.^{6-8,24,31-34)}

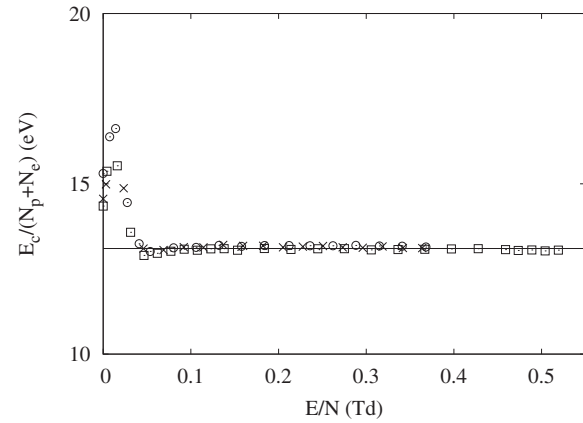


Fig. 11. Electric-field dependence of $N_p + N_e$ in xenon gas at 0.055 (○), 0.092 (□), and 0.124 g/cm³ (×). The solid line shows the average value above 0.05 Td (13.1 eV). The ionization charge uncollected by our main amplifier forms the peaks below 0.05 Td.

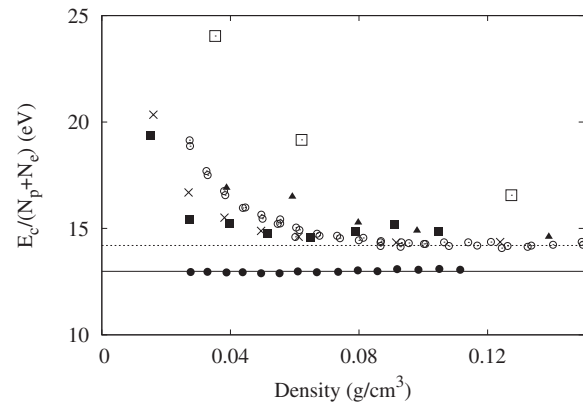


Fig. 12. Density dependence of $N_p + N_e$ in xenon gas at 0.4 Td (●). The solid line shows the average (13.0 eV). For comparison, the results of E_c/N_p measured at zero electric field using shaping amplifiers with time constants of 50 (○) and 300 μs (×) are shown. The dashed line shows the saturated value of E_c/N_p (14.2 eV). Also shown are previous relative scintillation yields reported by other groups: Dolgoshein *et al.*³⁸⁾ (□), Suzuki *et al.*¹¹⁾ (■), and Bolotnikov and Ramsey³⁰⁾ (▲). Those^{11,30,38)} are calibrated with our absolute scintillation yields.

Our result of W_{ex} -value measured with α -particles is well consistent with that of Saito *et al.*⁶⁾ at 0.005 g/cm³. In contrast, our W_{ex} -value is largely different from that of Saito *et al.*⁷⁾

We expect that the ratio of the scintillation yield originating from the excitation luminescence to total scintillation yield, measured by Saito *et al.*,⁷⁾ is excessively low because of the long STC (10 μs) of the main amplifier used in their work. In Miyajima *et al.*'s work,⁴⁾ the scintillation signals were shaped with a STC of 2 μs, which is shorter than the average thermalization time of e_{hot} (3 μs³⁵⁾ at 1 atm) in process (7). Most recombinations begin after the average thermalization time.³⁶⁾ Thus, recombination luminescence is not mostly collected within a collection time determined at a STC of 2 μs. In addition, $I(\lambda)$ assumed by Miyajima *et al.*⁴⁾ (~ 178 nm in peak³⁷⁾) is different from that assumed by us (~ 175 nm in peak⁵⁾). On both assumptions of no collection of recombination luminescence and of the

Table I. Comparison with previous studies in terms of W -value, W_{ex} -value, and N_{ex}/N_i .

Author	Density (g/cm ³)	Type ^{a)}	W_{ex} (eV)	W (eV)	N_{ex}/N_i
This work	0.028–0.111 (gas)	m	34.1 ± 2.4	20.9 ± 0.4	0.61 ± 0.04
Saito <i>et al.</i> ⁶⁾	0.005–0.056 (gas)	m	34.5 ^{b)}	20.9 ± 0.4	0.60
Saito <i>et al.</i> ⁷⁾	0.005–0.011 (gas)	m	59.4 ± 2.4	(22.1) ³²⁾	0.38 ± 0.02
Carmo <i>et al.</i> ⁸⁾	0.006 (gas)	m	111 ± 16	(21.61 ^{+0.14} _{-0.10}) ³³⁾	0.19 ^{d)}
Date <i>et al.</i> ²⁴⁾	0.001 (gas)	c	31.6 ^{c)}	21.7	0.69 ^{d)}
Doke <i>et al.</i> ³¹⁾	3.06 (liquid)	m	120 ^{d)}	(15.6 ± 0.3) ³⁴⁾	0.13

a) “m” and “c” represent measurement and calculation, respectively.

b) This value was determined from the average scintillation yield of 1.59×10^5 for 5.49 MeV α -particles.

c) This value was estimated from the average number of excitation collision events of 317 for 10 keV electrons. The average number was determined from the distribution of the frequency of excitation collisions shown in Fig. 7 in ref. 24.

d) These values are estimated by us from refs. 8, 24 and 31.

same $I(\lambda)$,⁵⁾ their value at 0.005 g/cm³ is estimated to be 46.5 eV. Taking account of the error of Q_{pmt} of their PMT that they neglected,⁴⁾ total error is estimated to be 19% owing to the steep curve of $q_e(\lambda)$ of their PMT in the wavelength range of 172–178 nm. The result of Miyajima *et al.*,⁴⁾ reanalyzed by us, is roughly in agreement with ours as shown in Fig. 7.

The Monte Carlo simulation of electron scattering processes performed in low-density xenon gas by Date *et al.*²⁴⁾ shows that the average number of excitation collision events for 10 keV electrons is 317, which corresponds to 31.6 eV in W_{ex} -value. Our experimental results are also roughly in agreement with the result of their Monte Carlo study. This means that the various data of ionization and excitation cross sections assumed by Date *et al.*²⁴⁾ are roughly correct, and additionally are applicable to high-density xenon gas up to 0.12 g/cm³.

According to Carmo *et al.*,⁸⁾ the W_{ex} -value for 5.9 keV X-rays is 111 ± 16 eV. We expect that the W_{ex} -value for such X-rays is nearly the same as that for the α -particles. This idea is supported by the calculation of Date *et al.*²⁴⁾ The difference between ours and theirs⁸⁾ is now under consideration with the result of Parsons *et al.*⁹⁾

5.2 Density dependence of total scintillation yield

In our previous work,¹³⁾ we measured relative scintillation yields at zero electric field using the main amplifier with a STC of 300 μ s longer than the present one of 50 μ s. These results were calibrated using the present results and plotted as E_c/N_p in Fig. 12. For comparison, previous results by Dolgoshein *et al.*³⁸⁾ and Suzuki *et al.*¹¹⁾ were also calibrated with a W_{ex} -value of 34.1 eV and plotted in Fig. 12. Furthermore, previous results according to Bolotnikov and Ramsey³⁰⁾ were calibrated with W_s -value at 0.21 g/cm³ in our previous work²²⁾ and plotted.

As shown in Fig. 8, our results of E_c/N_p observed at 50 μ s in STC, which are consistent with those of Saito *et al.*,⁵⁾ shows an increase in scintillation yield with density below 0.09 g/cm³. The increase in total scintillation yield above 0.005 g/cm³ is generally interpreted as an increase in recombination probability.³⁸⁾ However, it is shown in Fig. 12 that our results of E_c/N_p at 50 μ s are always higher than those at 300 μ s below 0.09 g/cm³. This means that recombination does not finish below 0.09 g/cm³ within the

collection time determined at a STC of 50 μ s. The results of E_c/N_p below 0.09 g/cm³ should be considered as the upper limit of W_s .

Our results of E_c/N_p at 300 μ s are lower than those according to Dolgoshein *et al.*,³⁸⁾ Bolotnikov and Ramsey,³⁰⁾ and Saito *et al.*⁵⁾ This is due to the longer collection time than theirs.^{5,30,38)} The result of E_c/N_p reported by Suzuki *et al.*,¹¹⁾ which has been obtained by measuring the current of PMT, seems to be lower than ours at 300 μ s. This may be caused by the longer collection time than ours. It is unclear whether or not the decrease in E_c/N_p with density, observed by Suzuki *et al.*,¹¹⁾ means an increase in recombination probability.

Above 0.09 g/cm³, the results of E_c/N_p at 50 μ s are well consistent with those at 300 μ s. The consistence indicates that the recombination finishes within the collection time. In addition, E_c/N_p at 50 μ s is saturated above 0.09 g/cm³. At the saturation point, the W_s -value for total scintillation yield is 14.2 ± 1.0 eV. The W_s -value is nearly consistent with the minimum W_s in liquid xenon (13.8 ± 0.9 eV).³¹⁾

As shown in Fig. 12, the saturated W_s -value of 14.2 eV at zero electric field is found to be higher than $E_c/(N_p + N_e)$ of 13.0 eV. This means that some $\text{Xe}_2^+ - e_{\text{th}}$ pairs do not recombine at zero electric field. The escape probability of e_{th} from Xe_2^+ is not mostly affected by xenon density above 0.09 g/cm³ and is calculated to be approximately 14%.

6. Conclusions

The average numbers of scintillation photons and electrons produced by an α -particle were measured in high-density xenon gas ranging from 0.03 to 0.12 g/cm³. Experimental results show that there is no significant density dependence of W_{ex} -value or W -value in the investigated density region. W_{ex} -value and W -value were determined to be 34.1 ± 2.4 and 20.9 ± 0.4 eV, respectively. It was also found that one recombined $\text{Xe}_2^+ - e_{\text{th}}$ pair produces one scintillation photon. In contrast, the total scintillation yield measured at zero electric field was found to be much lower than the sum of the scintillation yield and ionization yield at a high electric field (>0.05 Td). This means that some liberated electrons escape from recombination at zero electric field. The escape probability of liberated electrons is approximately 14% above 0.09 g/cm³.

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